

# 14

## Stereochemistry

### Connections

#### ➡ Building on

- Drawing organic molecules [ch2](#)
- Organic structures [ch4](#)
- Nucleophilic addition to the carbonyl group [ch6](#)
- Nucleophilic substitution at carbonyl groups [ch10](#) & [ch11](#)

#### Arriving at

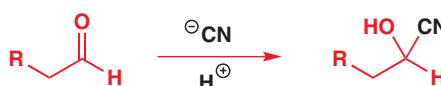
- Three-dimensional shape of molecules
- Molecules with mirror images
- Molecules with symmetry
- How to separate mirror-image molecules
- Diastereoisomers
- Shape and biological activity
- How to draw stereochemistry

#### ➡ Looking forward to

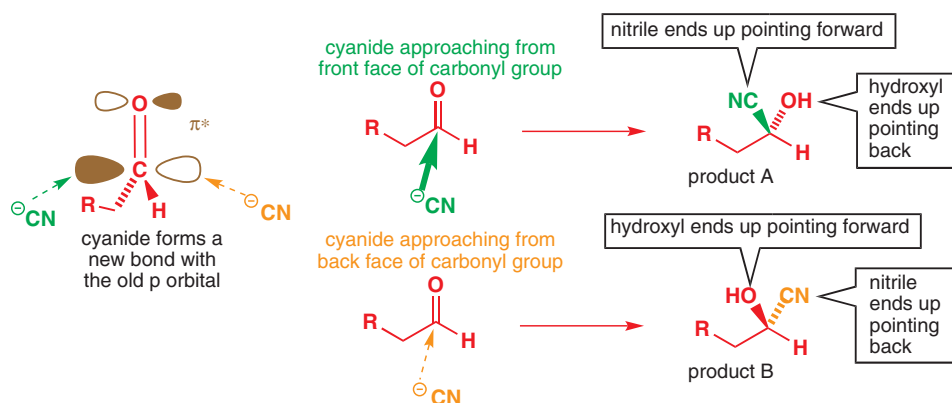
- Nucleophilic substitution at saturated C [ch15](#)
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### Some compounds can exist as a pair of mirror-image forms

One of the very first reactions you met, back in Chapter 6, was between an aldehyde and cyanide. The product was a compound containing a nitrile group and a hydroxyl group.



How many products are formed in this reaction? Well, the straightforward answer is one—there's only one aldehyde, only one cyanide ion, and only one reasonable way in which they can react. But this analysis is not *quite* correct. One point that we ignored when we first talked about this reaction, because it was irrelevant at that time, is that the carbonyl group of the aldehyde has two faces. The cyanide ion could attack either from the front face or the back face, giving, in each case, a distinct product.



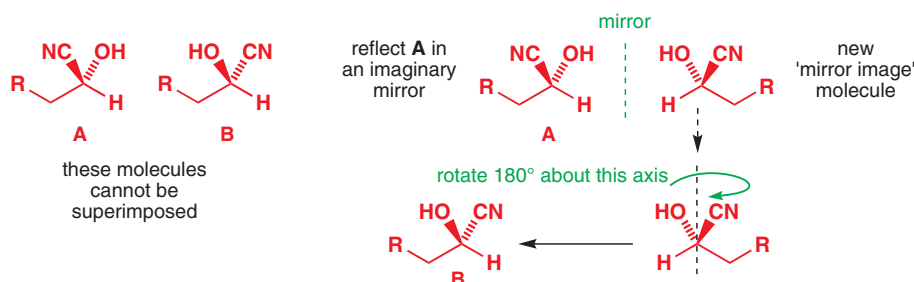
Interactive results of cyanide addition to carbonyls

■ The bold wedges represent bonds coming towards you, out of the paper, and the cross-hatched bonds represent bonds going away from you, into the paper.

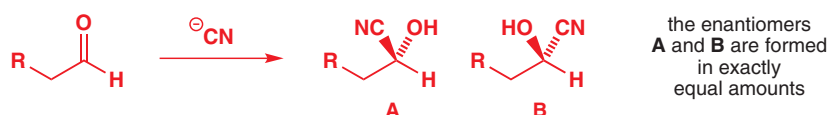
**Online support.** The icon in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type [www.chemtube3d.com/clayden/123](http://www.chemtube3d.com/clayden/123) into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

As we explained in Chapter 6 (pp. 125–7), the cyanide attacks the  $\pi^*$  orbital of the aldehyde more or less at right angles to the plane of the molecule as it forms a new bond with the old p orbital on C. This translates into ‘front’ and ‘back’ on a diagram on paper. Compare the diagram on the left with the others to make sure this is clear.

Are these two products different? If we lay them side by side and try to arrange them so that they look identical, we find that we can’t—you can verify this by making models of the two structures. The structures are non-superimposable—so they are not identical. In fact, they are mirror images of each other: if we reflected one of the structures, A, in a mirror, we would get a structure that is identical with B.



We call two structures that are not identical but are mirror images of each other (like these two) **enantiomers**. Structures that are not superimposable on their mirror image, and can therefore exist as two enantiomers, are called **chiral**. In this reaction, the cyanide ions are just as likely to attack the ‘front’ face of the aldehyde as they are the ‘back’ face, so we get a 50:50 mixture of the two enantiomers.



■ In reading this chapter, you will have to do a lot of mental manipulation of three-dimensional shapes. Because we can represent these shapes in the book only in two dimensions, we suggest that you make models, using a molecular model kit, of the molecules we talk about. With some practice, you will be able to imagine the molecules you see on the page in three dimensions.

Interactive aldehyde cyanohydrin structures—chiral

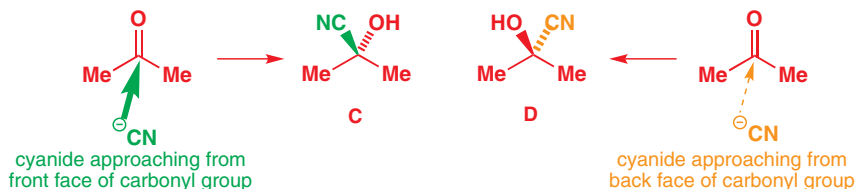
### ● Enantiomers and chirality

- Enantiomers are structures that are not identical, but are *mirror images* of each other.
- Structures are *chiral* if they cannot be superimposed on their mirror image.

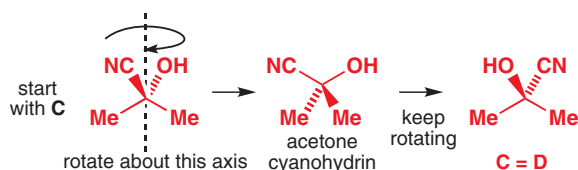
Now consider another similar reaction—the addition of cyanide to acetone.



Again an adduct (a cyanohydrin) is formed. You might imagine that attacking the front or the back face of the acetone molecule could again give two structures, C and D.



However, this time rotating one to match the other shows that they are superimposable and therefore identical.

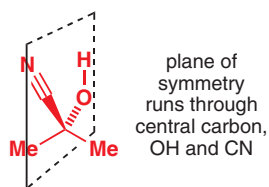


Interactive acetone cyanohydrin structure—achiral

Make sure that you are clear about this: **C** and **D** are identical molecules, while **A** and **B** are mirror images of each other. Reflection in a mirror makes no difference to **C** or **D**; they are superimposable on their own mirror images and therefore cannot exist as two enantiomers. Structures that are superimposable on their mirror images are called achiral.

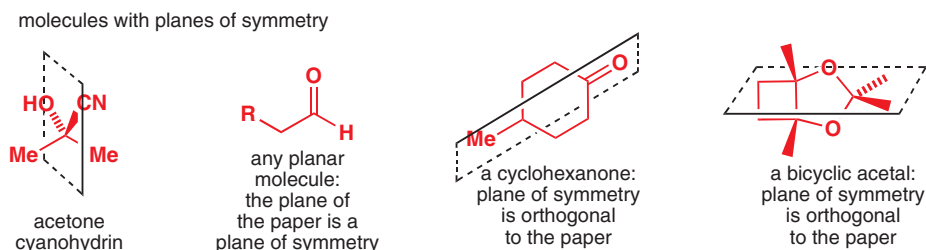
● **Achiral structures are superimposable on their mirror images.**

### Chiral molecules have no plane of symmetry

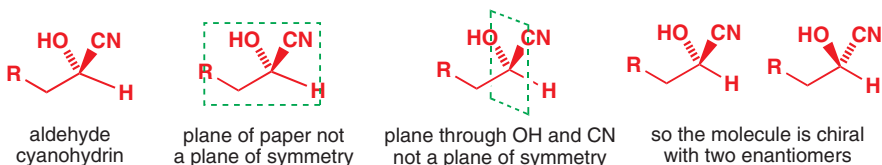


What is the essential difference between these two compounds that means one is superimposable on its mirror image and one is not? The answer is *symmetry*. Acetone cyanohydrin has a plane of symmetry running through the molecule. This plane cuts the central carbon and the OH and CN groups in half, and has one methyl group on each side. All planar molecules (such as our simple aldehyde) cannot be chiral as the plane of the molecule must be a plane of symmetry. Cyclic molecules may have a plane of symmetry passing through two atoms of the ring, as in the cyclohexanone below. The plane passes through both atoms of the carbonyl group and bisects the methyl group as well as the hydrogen atom (not shown) on the same carbon atom. The bicyclic acetal looks more complicated but a plane of symmetry passes between the two oxygen atoms and the two ring-junction carbon atoms while bisecting the two methyl groups. None of these molecules is chiral.

Interactive molecules with a plane of symmetry



On the other hand, the aldehyde cyanohydrin has no plane of symmetry: the plane of the paper has OH on one side and CN on the other while the plane at right angles to the paper has H on one side and RCH<sub>2</sub> on the other. This compound has no plane of symmetry (has *asymmetry*) and has two enantiomers.



Later in this chapter we shall meet a much less important type of symmetry that also means molecules are not chiral if they possess it. This is a centre of symmetry.

### Planes of symmetry and chirality

- Any structure that has no plane of symmetry is chiral and can exist as two mirror-image forms (*enantiomers*).
- Any structure with a plane of symmetry is not chiral and cannot exist as two enantiomers.

By 'structure', we don't just mean chemical structure: the same rules apply to everyday objects. Some examples from among more familiar objects in the world around us should help make these ideas clear. Look around you and find a chiral object—a pair of scissors, a screw (but not the screwdriver), a car, and anything with writing on it, like this page. Look again for achiral objects with planes of symmetry—a plain mug, saucepan, chair, most simple manufactured objects without writing on them. The most significant chiral object near you is the hand you write with.

### Gloves, hands, and socks



Most gloves exist in pairs of non-identical mirror-image forms: only a left glove fits a left hand and only a right glove fits a right hand. This property of gloves and of the hands inside them gives us the word 'chiral'—*cheir* is Greek for 'hand'. Hands and gloves are chiral; they have no plane of symmetry, and a left glove is not superimposable on its mirror image (a right glove). Feet are chiral too, as are shoes. But socks (usually!) are not. Although we all sometimes have problems finding two socks of a matching colour, once you've found them, you never have to worry about which sock goes on which foot because socks are achiral. A pair of socks is manufactured as two identical objects, each of which has a mirror plane.

The ancient Egyptians had less care for the chirality of hands and their paintings often show people, even Pharaohs, with two left hands or two right hands—they just didn't seem to notice.



### Tennis racquets and golf clubs



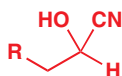
If you are left-handed and want to play golf, you either have to play in a right-handed manner or get hold of a set of left-handed golf clubs. Golf clubs are clearly therefore chiral; they can exist as either of two enantiomers. You can tell this just by looking at a golf club. It has no plane of symmetry, so it must be chiral. But left-handed tennis players have no problem using the same racquets as right-handed tennis players and tennis players of either chirality sometimes swap the racquet from hand to hand. Look at a tennis racquet: it has a plane of symmetry (indeed, usually two), so it's achiral. It can't exist as two mirror-image forms.

■ These statements are slightly incomplete but will serve you well in almost all situations: we will come to centres of symmetry shortly (p. 321).

### ● To summarize

- A structure *with* a plane of symmetry is *achiral* and *superimposable* on its mirror image and *cannot* exist as two enantiomers.
- A structure *without* a plane of symmetry is *chiral* and *not superimposable* on its mirror image and *can* exist as two enantiomers.

### Stereogenic centres

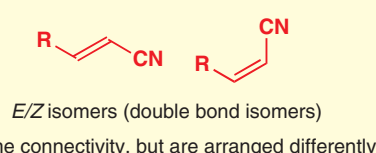
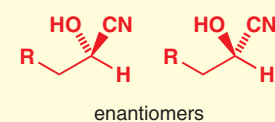
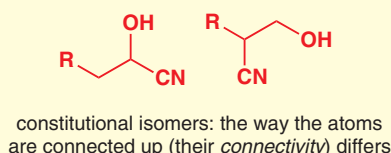


Back to chemistry, and the product from the reaction of an aldehyde with cyanide. We explained above that this compound, being chiral, can exist as two enantiomers. Enantiomers are clearly isomers; they consist of the same parts joined together in a different way. In particular, enantiomers are a type of isomer called **stereoisomers** because the isomers differ not in the connectivity of the atoms, but only in the overall shape of the molecule.

### ● Stereoisomers and constitutional isomers

Isomers are compounds that contain the same atoms bonded together in different ways. If the connectivity of the atoms in the two isomers is different, they are constitutional isomers. If the connectivity of the atoms in the two isomers is the same,

they are stereoisomers. Enantiomers are stereoisomers, and so are *E* and *Z* double bonds. We shall meet other types of stereoisomers shortly.



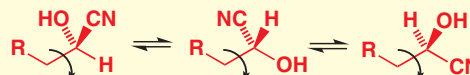
We should also introduce you briefly to another pair of concepts here, which you will meet again in more detail in Chapter 16: *configuration* and *conformation*. Two stereoisomers really are different molecules: they cannot be interconverted without breaking a bond somewhere. We therefore say that they have different configurations. But any molecule can exist in a number of conformations: two conformations differ only in the temporary way the molecule happens to arrange itself, and can easily be interconverted just by rotating around bonds. Humans all have the same *configuration*: two arms joined to the shoulders. We may have different *conformations*: arms folded, arms raised, pointing, waving, etc.

### ● Configuration and conformation

- Changing the *configuration* of a molecule always means that bonds are broken.
- A different configuration is a different molecule.
- Changing the *conformation* of a molecule means rotating about bonds, but not breaking them.
- Conformations of a molecule are readily interconvertible, and are all the same molecule.



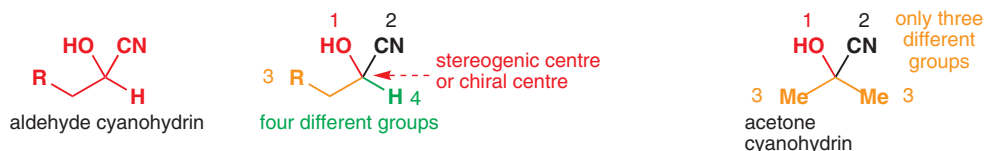
two configurations: going from one enantiomer to the other requires a bond to be broken



three conformations of the same enantiomer: getting from one to the other just requires rotation about a bond: all three are the same molecule

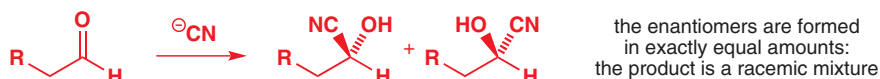
Interactive cyanohydrin conformations

The aldehyde cyanohydrin is chiral because it does not have a plane of symmetry. In fact, it *cannot* have a plane of symmetry because it contains a tetrahedral carbon atom carrying four different groups: OH, CN, RCH<sub>2</sub>, and H. Such a carbon atom is known as a **stereogenic** or **chiral centre**. The product of cyanide and acetone is not chiral; it has a plane of symmetry and no chiral centre because two of the groups on the central carbon atom are the same.



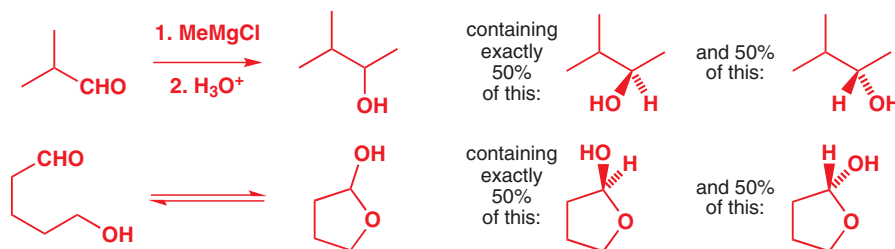
- If a molecule contains one carbon atom carrying four different groups it will not have a plane of symmetry and must therefore be chiral. A carbon atom carrying four different groups is a **stereogenic or chiral centre**.

We saw how the two enantiomers of the aldehyde cyanohydrin arose by attack of cyanide on the two faces of the carbonyl group of the aldehyde. We said that there was nothing to favour one face over the other, so the enantiomers must be formed in equal quantities. A mixture of equal quantities of a pair of enantiomers is called a **racemic mixture**.



- A **racemic mixture** is a mixture of two enantiomers in equal proportions. This principle is very important. If all the starting materials and reagents in a reaction are achiral and the products are chiral they will be formed as a racemic mixture of two enantiomers.

Here are some more reactions you have come across that make chiral products from achiral starting materials. In each case, the principle must hold—equal amounts of the two enantiomers (racemic mixtures) are formed.



### Many chiral molecules are present in nature as single enantiomers

Let's turn to some simple, but chiral, molecules—the natural amino acids. All amino acids have a carbon carrying an amino group, a carboxyl group, a hydrogen atom, and the R group, which varies from amino acid to amino acid. So unless R=H (this is the case for glycine), amino acids always contain a chiral centre and lack a plane of symmetry.



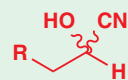
It is possible to make amino acids quite straightforwardly in the laboratory. The scheme below shows a synthesis of alanine, for example. It is a version of the Strecker synthesis you met in Chapter 11.

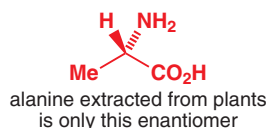
laboratory synthesis of racemic alanine from acetaldehyde



■ You will see shortly that compounds with *more than one* chiral centre are not always chiral.

■ When we don't show bold and dashed bonds to indicate the three-dimensional structure of the molecule, we mean that we are talking about both enantiomers of the molecule. Another useful way of representing this is with wiggly bonds. Wiggly bonds are in fact slightly ambiguous: here the wiggly bond means both stereoisomers. Elsewhere a wiggly bond might mean just one stereoisomer, but with unknown stereochemistry.





Alanine made in this way must be racemic because the starting material and all reagents are achiral. However, if we isolate alanine from a natural source—by hydrolysing vegetable protein, for example—we find that this is not the case. Natural alanine is solely one enantiomer, the one drawn in the margin. Samples of chiral compounds that contain only one enantiomer are called **enantiomerically pure**. We know that ‘natural’ alanine contains only this enantiomer from X-ray crystal structures.

### Enantiomeric alanine

In fact, nature does sometimes (but very rarely) use the other enantiomer of alanine, for example in the construction of bacterial cell walls. Some antibiotics (such as vancomycin) owe their selectivity to the way they can recognize these ‘unnatural’ alanine components and destroy the cell wall that contains them.

### Chiral and enantiomerically pure

Before we go further, we should just mention one common point of confusion. Any compound whose molecules do not have a plane of symmetry is chiral. Any sample of a chiral compound that contains molecules all of the same enantiomer is enantiomerically pure. *All* alanine is chiral (the structure has no plane of symmetry) but *laboratory-produced* alanine is racemic (a 50:50 mixture of enantiomers) whereas *naturally isolated* alanine is enantiomerically pure.

#### ● ‘Chiral’ does not mean ‘enantiomerically pure’.

Most of the molecules we find in nature are chiral—a complicated molecule is much more likely not to have a plane of symmetry than to have one. Nearly all of these chiral molecules in living systems are found not as racemic mixtures, but as single enantiomers. This fact has profound implications, for example in the chemistry of drug design, and we will come back to it later.

### R and S can be used to describe the configuration of a chiral centre

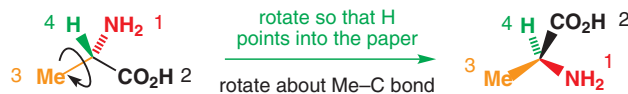
Before going on to talk about single enantiomers of chiral molecules in more detail, we need to explain how chemists describe which enantiomer they’re talking about. We can, of course, just draw a diagram, showing which groups go into the plane of the paper and which groups come out of the plane of the paper. This is best for complicated molecules. Alternatively, we can use the following set of rules to assign a letter, *R* or *S*, to describe the configuration of groups at a chiral centre in the molecule.

Here again is the enantiomer of alanine you get if you extract alanine from living things.

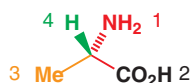
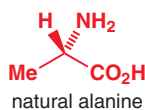
1. Assign a priority number (1–4) to each substituent at the chiral centre. Atoms with higher atomic numbers get higher priority.

Alanine’s chiral centre carries one N atom (atomic number 7), two C atoms (atomic number 6), and one H atom (atomic number 1). So, we assign priority 1 to the  $\text{NH}_2$  group, because N has the highest atomic number. Priorities 2 and 3 will be assigned to the  $\text{CO}_2\text{H}$  and the  $\text{CH}_3$  groups, and priority 4 to the hydrogen atom; but we need a way of deciding which of  $\text{CO}_2\text{H}$  and  $\text{CH}_3$  takes priority over the other. If two (or more) of the atoms attached to the chiral centre are identical, then we assign priorities to these two by assessing the atoms attached to those atoms. In this case, one of the carbon atoms carries oxygen atoms (atomic number 8) and one carries only hydrogen atoms (atomic number 1). So  $\text{CO}_2\text{H}$  is higher priority than  $\text{CH}_3$ ; in other words,  $\text{CO}_2\text{H}$  gets priority 2 and  $\text{CH}_3$  priority 3.

2. Arrange the molecule so that the lowest priority substituent is pointing away from you. In our example, naturally extracted alanine, H is priority 4, so we need to look at the molecule with the H atom pointing into the paper, like this.



■ Remember—we use the word *configuration* to describe the arrangement of bonds around an atom. Configurations cannot be changed without breaking bonds.



Interactive configuration assignment

■ These priority rules are also used to assign *E* and *Z* to alkenes, and are sometimes called the Cahn–Ingold–Prelog (CIP) rules, after their devisors. You can alternatively use atomic weights—for isotopes you have to (D has a higher priority than H)—apart from in the vanishingly rare case of a chiral centre bearing Te and I (look at the data in the periodic table at the front of this book to see why).



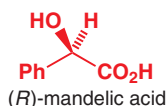
The angle through which a sample of a compound (usually a solution) rotates plane-polarized light depends on a number of factors, the most important ones being the path length (how far the light has to pass through the solution), concentration, temperature, solvent, and wavelength. Typically, optical rotations are measured at 20 °C in a solvent such as ethanol or chloroform, and the light used is from a sodium lamp, with a wavelength of 589 nm.

The observed angle through which the light is rotated is given the symbol  $\alpha$ . By dividing this value by the path length  $l$  (in dm) and the concentration  $c$  (in g cm<sup>-3</sup>) we get a value,  $[\alpha]$ , which is specific to the compound in question. Indeed,  $[\alpha]$  is known as the compound's **specific rotation**. The choice of units is eccentric and arbitrary but is universal so we must live with it.

$$[\alpha] = \frac{\alpha}{cl}$$

Most  $[\alpha]$  values are quoted as  $[\alpha]_D$  (where the D indicates the wavelength of 589 nm, the 'D line' of a sodium lamp) or  $[\alpha]_D^{20}$ , the 20 indicating 20 °C. These define the remaining variables.

Here is an example. A simple acid, known as mandelic acid, can be obtained from almonds in an enantiomerically pure form. When 28 mg was dissolved in 1 cm<sup>3</sup> of ethanol and the solution placed in a 10-cm-long polarimeter cell, an optical rotation  $\alpha$  of  $-4.35^\circ$  was measured (that is,  $4.35^\circ$  to the left) at 20 °C with light of wavelength 589 nm. What is the specific rotation of the acid?



■ Note that the units of a measured optical rotation  $\alpha$  are degrees, but, by convention, specific rotation  $[\alpha]$  is quoted without units.

First, we need to convert the concentration to grammes per cubic centimetre: 28 mg in 1 cm<sup>3</sup> is the same as 0.028 g cm<sup>-3</sup>. The path length of 10 cm is 1 dm, so

$$[\alpha]_D^{20} = \frac{\alpha}{cl} = \frac{-4.35}{0.028 \times 1} = -155.4$$

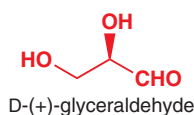
### Enantiomers can be described as (+) or (–)

We can use the fact that two enantiomers rotate plane-polarized light in opposite directions to assign each a label that doesn't depend on knowing its configuration. We call the enantiomer that rotates plane-polarized light to the right (gives a positive rotation) the (+)-enantiomer (or the *dextrorotatory* enantiomer) and the enantiomer that rotates plane-polarized light to the left (gives a negative rotation) the (–)-enantiomer (or the *laevorotatory* enantiomer). The direction in which light is rotated is not dependent on whether a stereogenic centre is R or S. An (R) compound is equally as likely to be (+) as (–)—of course, if it is (+) then its (S) enantiomer must be (–). The enantiomer of mandelic acid we have just discussed, for example, is (R)-(–)-mandelic acid because its specific rotation is negative, and (S)-alanine happens to be (S)-(+)-alanine. The labels (+) and (–) were more useful before the days of X-ray crystallography, when chemists did not know the actual configuration of the molecules they studied, and could distinguish two enantiomers only by the signs of their specific rotations.

### Enantiomers can be described as D or L

Long before the appearance of X-ray crystallography as an analytical tool, chemists had to discover the detailed structure and stereochemistry of molecules by a complex series of degradations. A molecule was gradually broken down into its constituents, and from the products that were formed the overall structure of the starting molecule was deduced. As far as stereochemistry was concerned, it was possible to measure the specific rotation of a compound, but not to determine its configuration. However, by using series of degradations it was possible to tell whether certain compounds had the same or opposite configurations.

Glyceraldehyde is one of the simplest chiral compounds in nature. Because of this, chemists took it as the standard against which the configurations of other compounds could be compared. The two enantiomers of glyceraldehyde were given the labels D (for dextro—because it was the (+)-enantiomer) and L (for laevo—because it was the (–)-enantiomer). Any enantiomerically pure compound that could be related, by a series of chemical degradations and transformations, to D-(+)-glyceraldehyde was labelled D, and any compound that could be

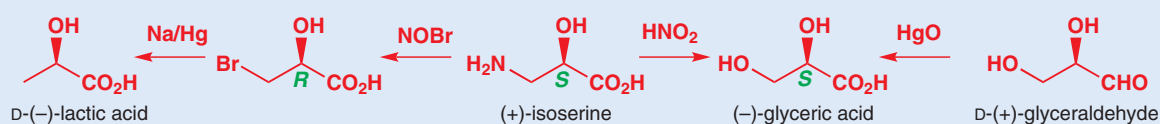


related to L-(–)-glyceraldehyde was labelled L. The processes concerned were slow and laborious (the scheme below shows how (–)-lactic acid was shown to be D-(–)-lactic acid) and are never used today. D and L are now used only for certain well-known natural molecules, where their use is established by tradition, for example, the L-amino acids or the D-sugars. These labels, D and L, are in *small capital* letters.

- Remember that the *R/S*, *+/-*, and *D/L* nomenclatures all arise from different observations and the fact that a molecule has, say, the *R* configuration gives no clue as to whether it will have *+* or *–* optical activity or be labelled D or L. Never try and label a molecule as *D/L* or *+/-* simply by working it out from the structure. Likewise, never try and predict whether a molecule will have a *+* or *–* specific rotation by looking at the structure.

### The correlation between D-(–)-lactic acid and D-(+)-glyceraldehyde

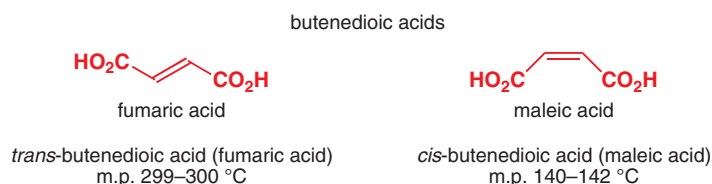
Here, for example, is the way that (–)-lactic acid was shown to have the same configuration as D-(+)-glyceraldehyde. We do not expect you to have come across the reactions used here.



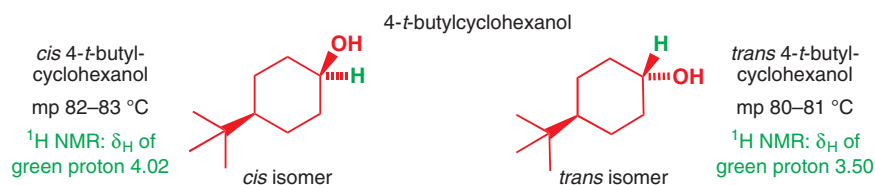
Notice from this scheme that the three intermediates all have the 'same' stereochemistry and yet one is (*R*) and two are (*S*). This is merely the consequence of the priority of the elements. (*R*) can be D or L and (*+*) or (*–*).

## Diastereoisomers are stereoisomers that are not enantiomers

Two enantiomers are chemically identical because they are mirror images of one another. Other types of stereoisomers may be chemically (and physically) quite different. These two alkenes, for example, are geometrical isomers (or *cis-trans* isomers). Their physical chemical properties are different, as you would expect, since they are quite different in shape. Neither is chiral of course as they are planar molecules.



A similar type of stereoisomerism can exist in cyclic compounds. In one of these, 4-*t*-butylcyclohexanols, the two substituents are on the same side of the ring; in the other, they are on opposite sides of the ring. Again, the two compounds have chemical and physical properties that are quite different.



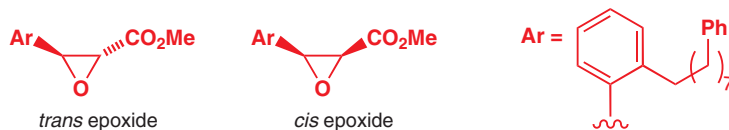
■ Notice that we do not always write in all the hydrogen atoms. If the *t*-butyl group is forward, as in these diagrams, the hydrogen atom must be back.

Stereoisomers that are not mirror images of one another are called *diastereoisomers*. Both of these pairs of isomers fall into this category. Notice how the physical and chemical properties of a pair of diastereoisomers differ.

- The physical and chemical properties of enantiomers are identical; the physical and chemical properties of diastereoisomers differ. 'Diastereoisomer' is sometimes shortened to 'diastereomer'.

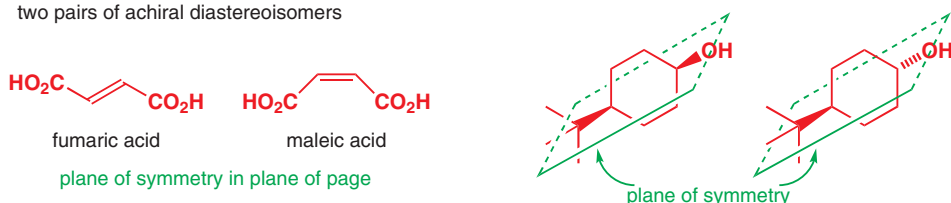
### Diastereoisomers can be chiral or achiral

This pair of epoxides was produced by chemists in Pennsylvania in the course of research on drugs intended to alleviate the symptoms of asthma. Clearly, they are again diastereoisomers, and again they have different properties. Although the reaction they were using to make these compounds gave some of each diastereoisomer, the chemists working on these compounds only wanted to use the first (*trans*) epoxide. They were able to separate it from its *cis* diastereoisomer by chromatography because the diastereoisomers differ in polarity.



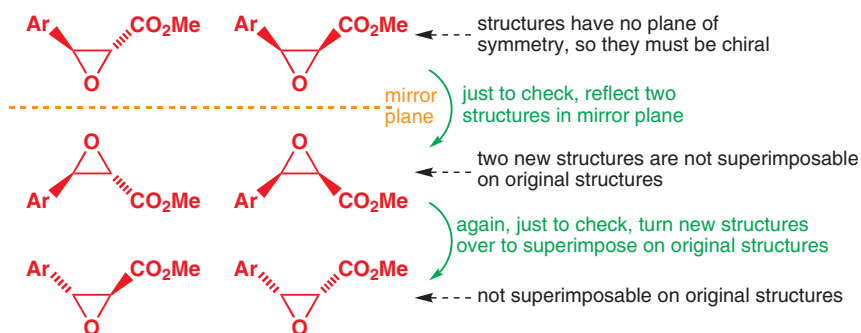
This time, the diastereoisomers are a little more complex than the examples above. The first two pairs of diastereoisomers we looked at were achiral—they each had a plane of symmetry through the molecule.

two pairs of achiral diastereoisomers

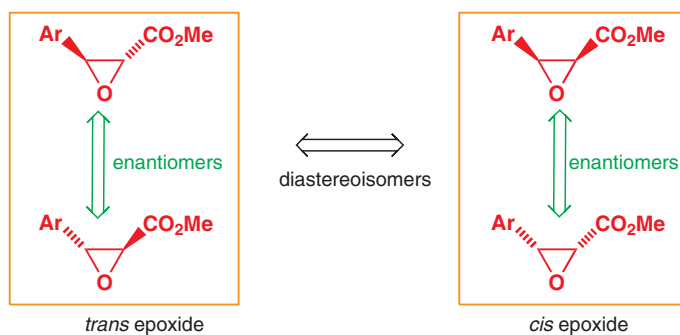


Interactive structures of achiral diastereoisomers

The epoxide diastereoisomers, on the other hand, are chiral. We know this because they do not have a plane of symmetry and we can check that by drawing the mirror image of each one: it is not superimposable on the first structure.



If a compound is chiral, it can exist as two enantiomers. We've just drawn the two enantiomers of each of the diastereoisomers of our epoxide. This set of four structures contains two diastereoisomers (stereoisomers that are not mirror images). These are the two different chemical compounds, the *cis* and *trans* epoxides, that have different properties. Each can exist as two enantiomers (stereoisomers that are mirror images) indistinguishable except for rotation. We have two pairs of diastereoisomers, each being a pair of enantiomers. When you are considering the stereochemistry of a compound, always distinguish the diastereoisomers first and then split these into enantiomers if they are chiral.



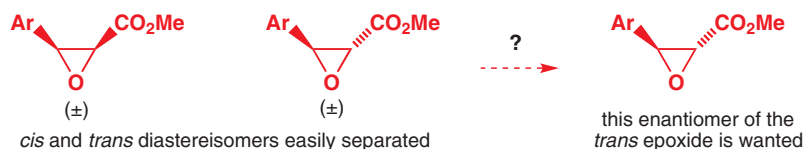
Interactive structures of epoxide diastereoisomers

In fact, the chemists working on these compounds wanted only one enantiomer of the *trans* epoxide—the top left stereoisomer. They were able to separate the *trans* epoxide from the *cis* epoxide by chromatography because they are diastereoisomers. However, because they had made both diastereoisomers in the laboratory from achiral starting materials, both diastereoisomers were racemic mixtures of the two enantiomers. Separating the top enantiomer of the *trans* epoxide from the bottom one was much harder because enantiomers have identical physical and chemical properties. To get just the enantiomer they wanted the chemists had to develop some completely different chemistry, using enantiomerically pure compounds derived from nature.

➡ We shall discuss how chemists make enantiomerically pure compounds later in this chapter, and in more detail in Chapter 41.

## Absolute and relative stereochemistry

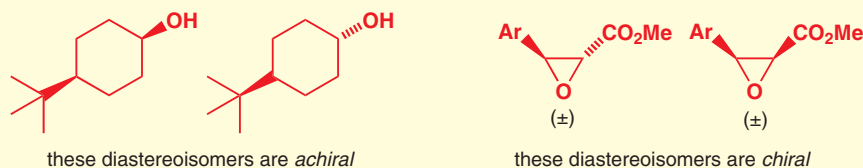
When we talk about two chiral diastereoisomers, we have no choice but to draw the structure of one enantiomer of each diastereoisomer because we need to include the stereochemical information to distinguish them, even if we're talking about a racemic mixture of the two enantiomers. To avoid confusion, it's best to write something definite under the structure, such as '±' (meaning racemic) under a structure if it means 'this diastereoisomer' but not 'this enantiomer of this diastereoisomer'. So in this case we should say that the chemists were able to separate these two diastereoisomers, but wanted only one enantiomer of the *trans* diastereoisomer and that this could not be separated by physical means.



When the stereochemistry drawn on a molecule means 'this diastereoisomer', we say that we are representing **relative stereochemistry**; when it means 'this enantiomer of this diastereoisomer' we say we are representing its **absolute stereochemistry**. Relative stereochemistry tells us only how the stereogenic centres *within a molecule* relate to each other.

### ● Enantiomers and diastereoisomers

- **Enantiomers** are stereoisomers that are mirror images. A pair of enantiomers are mirror-image forms of the same compound and have opposite **absolute stereochemistry**.
- **Diastereoisomers** are stereoisomers that are not mirror images. Two diastereoisomers are different compounds, and have different **relative stereochemistry**.
- Diastereoisomers may be achiral (have a plane of symmetry) or they may be chiral (have no plane of symmetry).

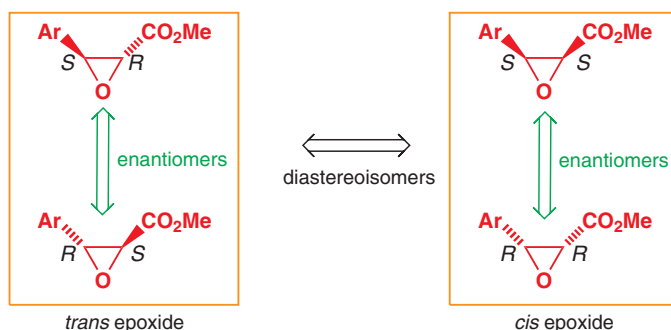


## Diastereoisomers can arise when structures have more than one stereogenic centre

Let's analyse our set of four stereoisomers a little more closely. You may have already noticed that these structures all contain stereogenic centres—two in each case. Go back to the diagram of the four structures at the bottom of p. 312 and, without looking at the structures overleaf, assign an *R* or *S* label to each of the stereogenic centres.

You should have made assignments of *R* and *S* like this.

➡ You need to know, and be able to use, the rules for assigning *R* and *S*; they were explained on p. 308. If you get any of the assignments wrong, make sure you understand why.

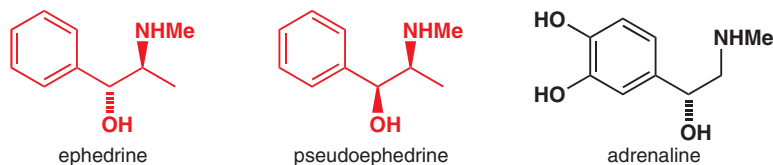


### ● Converting enantiomers and diastereoisomers

- To go from one *enantiomer* to another, *both* stereogenic centres are inverted.
- To go from one *diastereoisomer* to another, only *one* of the two is inverted.

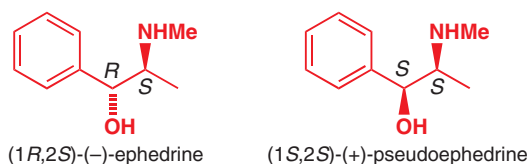
■ If you are asked to explain some stereochemical point in an examination, choose a cyclic example—it makes it much easier.

All the compounds that we have talked about so far have been cyclic because the diastereoisomers are easy to visualize: two diastereoisomers can be identified because the substituents are either on the same side or on opposite sides of the ring (*cis* or *trans*). But acyclic compounds can exist as diastereoisomers too. Take these two, for example. Both ephedrine and pseudoephedrine are members of the amphetamine class of stimulants, which act by imitating the action of the hormone adrenaline.



Ephedrine and pseudoephedrine are stereoisomers that are clearly not mirror images of each other—only one of the two stereogenic centres in ephedrine is inverted in pseudoephedrine—so they must be diastereoisomers. Thinking in terms of stereogenic centres is useful because, just as this compound has two stereogenic centres and can exist as two diastereoisomers, any compound with more than one stereogenic centre can exist in more than one diastereoisomeric form.

Both ephedrine and pseudoephedrine are produced in enantiomerically pure form by plants, so, unlike the anti-asthma intermediates above, in this case we are talking about single enantiomers of single diastereoisomers. Adrenaline (also known as epinephrine) is also chiral. In nature it is a single enantiomer but it cannot exist as other diastereoisomers as it has only one stereogenic centre.



### Ephedrine and pseudoephedrine

Ephedrine is a component of the traditional Chinese remedy 'Ma Huang', extracted from *Ephedra* species. It is also used in nasal sprays as a decongestant. Pseudoephedrine is the active component of the decongestant Sudafed.

The 'natural' enantiomers of the two diastereomers are (–)-ephedrine and (+)-pseudoephedrine, which does not tell you which is which, or (1*R*,2*S*)-(–)-ephedrine and (1*S*,2*S*)-(+)-

pseudoephedrine, which does. From that you should be able to deduce the corresponding structures.

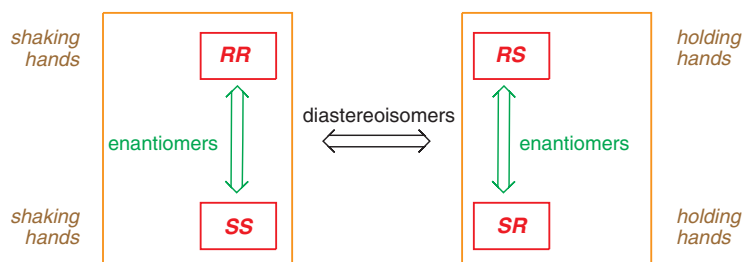
Here are some data on (1*R*,2*S*)-(-)-ephedrine and (1*S*,2*S*)-(+)-pseudoephedrine and their 'unnatural' enantiomers (which have to be made in the laboratory), (1*S*,2*R*)-(+)-ephedrine and (1*R*,2*R*)-(-)-pseudoephedrine.

|                                         | (1 <i>R</i> ,2 <i>S</i> )-(-)-ephedrine | (1 <i>S</i> ,2 <i>R</i> )-(+)-ephedrine | (1 <i>S</i> ,2 <i>S</i> )-(+)-pseudoephedrine | (1 <i>R</i> ,2 <i>R</i> )-(-)-pseudoephedrine |
|-----------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------------|-----------------------------------------------|
| mp                                      | 40–40.5 °C                              | 40–40.5 °C                              | 117–118 °C                                    | 117–118 °C                                    |
| [ $\alpha$ ] <sub>D</sub> <sup>20</sup> | –6.3                                    | +6.3                                    | +52                                           | –52                                           |

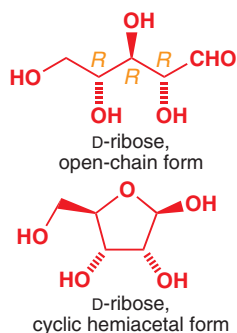
Remember that (+) and (–) refer to the sign of the specific rotation, while *R* and *S* are derived simply by looking at the structure of the compounds. There is no simple connection between the two!

● The two diastereoisomers are different compounds with different names and different properties, while the pairs of enantiomers are the same compound with the same properties, differing only in the direction in which they rotate polarized light.

We can illustrate the combination of two stereogenic centres in a compound by considering what happens when you shake hands with someone. Hand-shaking is successful only if you each use the same hand! By convention, this is your right hand, but it's equally possible to shake left hands. The overall pattern of interaction between two right hands and two left hands is the same: a right-handshake and a left-handshake are enantiomers of one another; they differ only in being mirror images. If, however, you misguidedly try to shake your right hand with someone else's left hand you end up holding hands. Held hands consist of one left and one right hand; a pair of held hands have totally different interactions from a pair of shaking hands; we can say that holding hands is a diastereoisomer of shaking hands. We can summarize the situation when we have two hands, or two chiral centres, each one *R* or *S*.



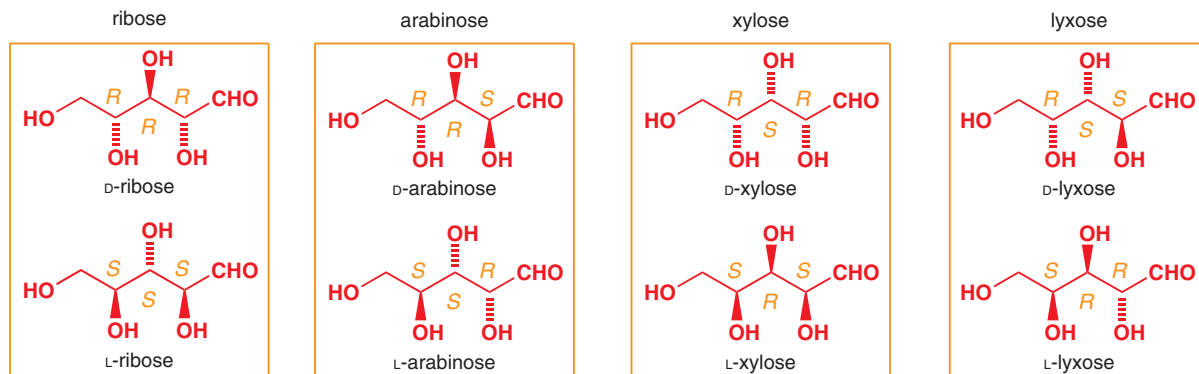
What about compounds with more than two stereogenic centres? The family of sugars provides lots of examples. Ribose is a five-carbon sugar that contains three stereogenic centres. The enantiomer shown here is the one used in the metabolism of all living things and, by convention, is known as D-ribose. The three stereogenic centres of D-ribose have the *R* configuration. For convenience, we will consider ribose in its open-chain form, but more usually it would be cyclic, as shown underneath.



In theory we can work out how many 'stereoisomers' there are of a compound with three stereogenic centres simply by noting that there are  $8 (= 2^3)$  ways of arranging Rs and Ss.

|            |            |            |            |
|------------|------------|------------|------------|
| <i>RRR</i> | <i>RRS</i> | <i>RSR</i> | <i>RSS</i> |
| <i>SSS</i> | <i>SSR</i> | <i>SRS</i> | <i>SRR</i> |

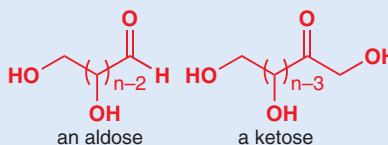
But this method blurs the all-important distinction between diastereoisomers and enantiomers. In each case, the combination in the top row and the combination directly below it are enantiomers (all three centres are inverted); the four columns are diastereoisomers. Three stereogenic centres therefore give four diastereoisomers, each a pair of two enantiomers. Going back to the example of the  $C_5$  aldoses, each of these diastereoisomers is a different sugar. In these diagrams each diastereoisomer is in a frame but the top line shows one enantiomer (D) and the bottom line the other (L).



■ You do not need to remember the names of these sugars.

### Structure of sugars

A sugar has the empirical formula  $C_nH_{2n}O_n$ , and consists of a chain of carbon atoms, one being a carbonyl group and the rest carrying OH groups. If the carbonyl group is at the end of the chain (in other words, it is an aldehyde), the sugar is an aldose. If the carbonyl group is not at the end of the chain, the sugar is a ketose. We come back to all this in detail in Chapter 42. The number of carbon atoms,  $n$ , can be 3–8: aldoses have  $n - 2$  stereogenic centres and ketoses  $n - 3$  stereogenic centres. In fact, most sugars exist as an equilibrium mixture of this open-chain structure and a cyclic hemiacetal isomer (Chapter 6).

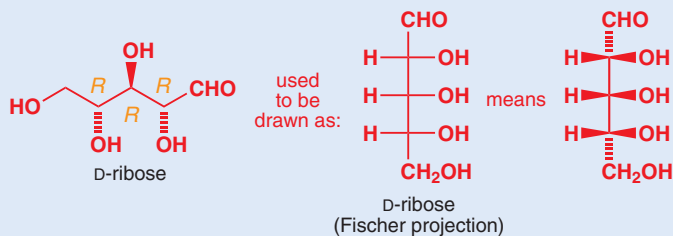


■ This is an oversimplification to be used cautiously because it works only if all diastereoisomers are chiral and fails with the sort of symmetrical molecules we are about to describe.

You've probably recognized that there's a simple mathematical relationship between the number of stereogenic centres and the number of stereoisomers a structure can have. **Usually**, a structure with  $n$  stereogenic centres can exist as  $2^n$  stereoisomers. These stereoisomers consist of  $2^{n-1}$  diastereoisomers, each of which has a pair of enantiomers.

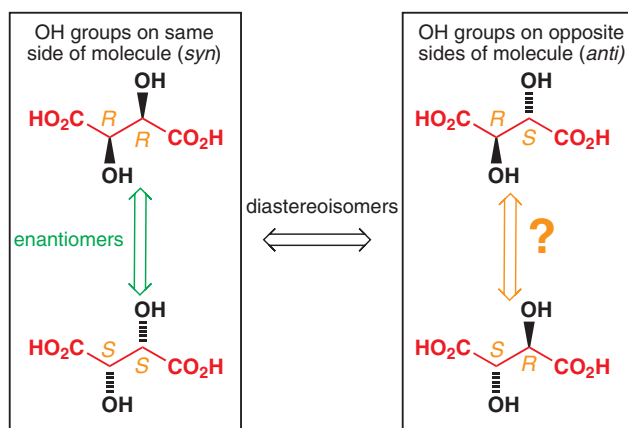
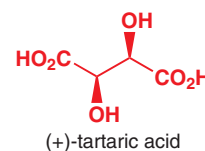
### Fischer projections

The stereochemistry of sugars used to be represented by Fischer projections. The carbon backbone was laid out in a vertical line and twisted in such a way that all the substituents pointed towards the viewer. Fischer projections are so unlike real molecules that you should never use them. However, you may see them in older books, and you should have an idea about how to interpret them. Just remember that all the branches down the side of the central trunk are effectively bold wedges (coming towards the viewer), while the central trunk lies in the plane of the paper. By mentally twisting the backbone into a realistic zig-zag shape you should end up with a reasonable representation of the sugar molecule.



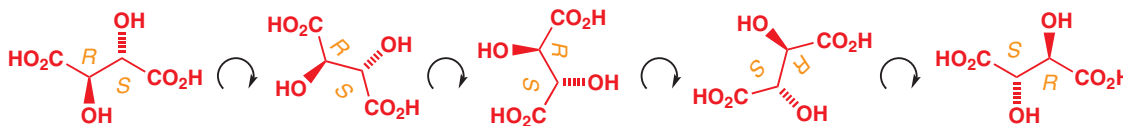
### Why only *usually*?—achiral compounds with more than one stereogenic centre

Sometimes, symmetry in a molecule can cause some stereoisomers to be degenerate, or ‘cancel out’—there aren’t as many stereoisomers as you’d *expect*. Take tartaric acid, for example. This stereoisomer of tartaric acid is found in grapes, and its salt, potassium hydrogen tartrate, can precipitate out as crystals at the bottom of bottles of wine. It has two stereogenic centres, so you’d expect  $2^2 = 4$  stereoisomers; two diastereoisomers, each a pair of enantiomers.



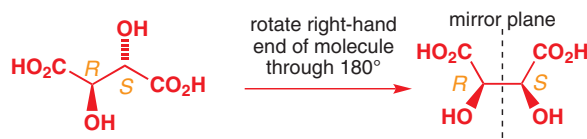
Interactive stereoisomers of tartaric acid

While the pair of structures on the left are certainly enantiomers, if you look carefully at the pair of structures on the right, you’ll see that they are, in fact, not enantiomers but identical structures. To prove it, just rotate the top one through  $180^\circ$  in the plane of the paper.



(1*R*,2*S*)-Tartaric acid and (1*S*,2*R*)-tartaric acid are not enantiomers, but they are identical because, even though they contain stereogenic centres, they are achiral. By drawing (1*R*,2*S*)-tartaric acid after a  $180^\circ$  rotation about the central bond, you can easily see that it has a mirror plane, and so must be achiral. Since the molecule has a plane of symmetry, and *R* is the mirror image of *S*, the *R,S* diastereoisomer cannot be chiral.

Interactive display of meso form of tartaric acid



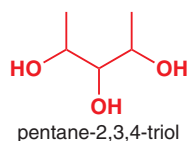
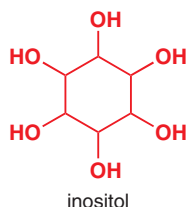
These two structures are the same molecule drawn in two different conformations—to get from one to the other just rotate half of the molecule about the central bond.

• Compounds that contain stereogenic centres but are themselves achiral are called **meso compounds**. This means that there is a plane of symmetry with *R* stereochemistry on one side and *S* stereochemistry on the other.

So tartaric acid can exist as two diastereoisomers, one with two enantiomers and the other achiral (a *meso* compound). It’s worth noting that the formula stating that a compound with  $n$  stereogenic centres has  $2^{n-1}$  diastereoisomers has worked but not the formula that states there are  $2^n$  ‘stereoisomers’. In general, it’s safer not to count up total ‘stereoisomers’ but to work out first how many diastereoisomers there are, and then to decide whether or not each one is chiral, and therefore whether or not it has a pair of enantiomers.

#### Meso hand-shaking

We can extend our analogy between hand-shaking and diastereoisomers to *meso* compounds as well. Imagine a pair of identical twins shaking hands. They could be shaking their left hands or their right hands and there would be a way to tell the two handshakes apart because they are enantiomers. But if the twins hold hands, you will not be able to distinguish left holds right from right holds left, because the twins themselves are indistinguishable—this is the *meso* hand-hold!



■ **syn and anti:** These refer to substituents on the same side (*syn*) or on opposite (*anti*) sides of a chain or ring. They must be used only in reference to a diagram.

|                   | Chiral diastereoisomer |                   | Achiral diastereoisomer |
|-------------------|------------------------|-------------------|-------------------------|
|                   | (+)-tartaric acid      | (-)-tartaric acid | meso-tartaric acid      |
| $[\alpha]_D^{20}$ | +12                    | -12               | 0                       |
| m.p.              | 168–170 °C             | 168–170 °C        | 146–148 °C              |

### Meso diastereoisomers of inositol

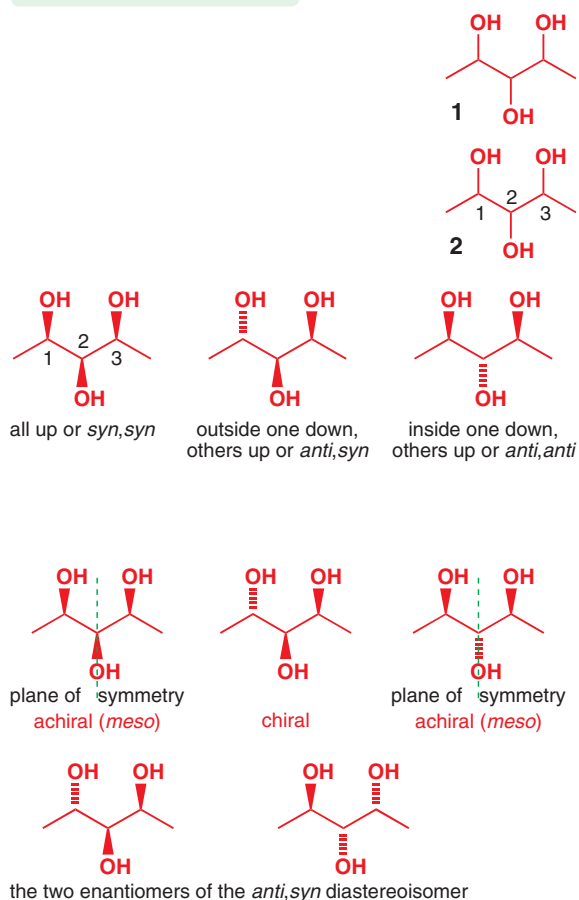
Look out for *meso* diastereoisomers in compounds that have a degree of symmetry in their overall structure. Inositol, one of whose diastereoisomers is an important growth factor, has six stereogenic centres. It's a challenge to work out how many diastereoisomers it has—in fact all but one of them are *meso*.

### Investigating the stereochemistry of a compound

When you want to describe the stereochemistry of a compound our advice is to identify the diastereoisomers and then think about whether they are chiral or not. Don't just count up 'stereoisomers'—to say that a compound with two stereogenic centres has four 'stereoisomers' is rather like saying that 'four hands are getting married'. Two people are getting married, each with two hands.

Let's work through how you might think about the stereochemistry of a simple example, the linear triol 2,3,4-trihydroxypentane or pentane-2,3,4-triol.

This is what you should do.



1. Draw the compound with the carbon skeleton in the usual zig-zag fashion running across the page, 1.

2. Identify the chiral centres, 2.

3. Decide how many diastereoisomers there are by putting the substituents at those centres up or down. It often helps to give each diastereoisomer a 'tag' name. In this case there are three diastereoisomers. The three OH groups can be all on the same side or else one of the end OHs or the middle one can be on the opposite side to the rest. We can call the first *syn,syn* because the two pairs of chiral centres (1 & 2, and 2 & 3) groups are both arranged with the OHs on the same side of the molecule (*syn*).

4. By checking on possible planes of symmetry, see which diastereoisomers are chiral. In this case only the plane down the centre can be a plane of symmetry.

5. Draw the enantiomers of any chiral diastereoisomer by inverting *all* the stereogenic centres. This can easily be achieved by reflecting the molecule in the plane of the paper, as if it were a mirror. Everything that was 'up' is now 'down' and vice versa.

6. Announce the conclusion. You could have said that there are four 'stereoisomers' but the following statement is much more helpful. There are three diastereoisomers, the *syn,syn*, the *syn,anti*, and the *anti,anti*. The *syn,syn* and the *anti,anti* are achiral (*meso*) compounds but the *syn,anti* is chiral and has two enantiomers.

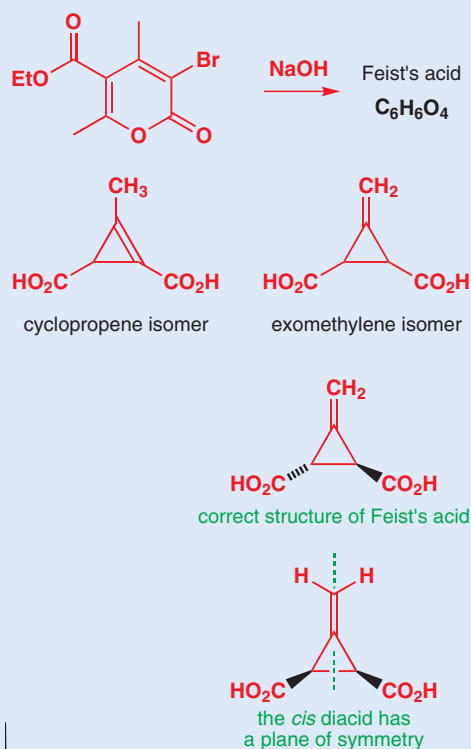
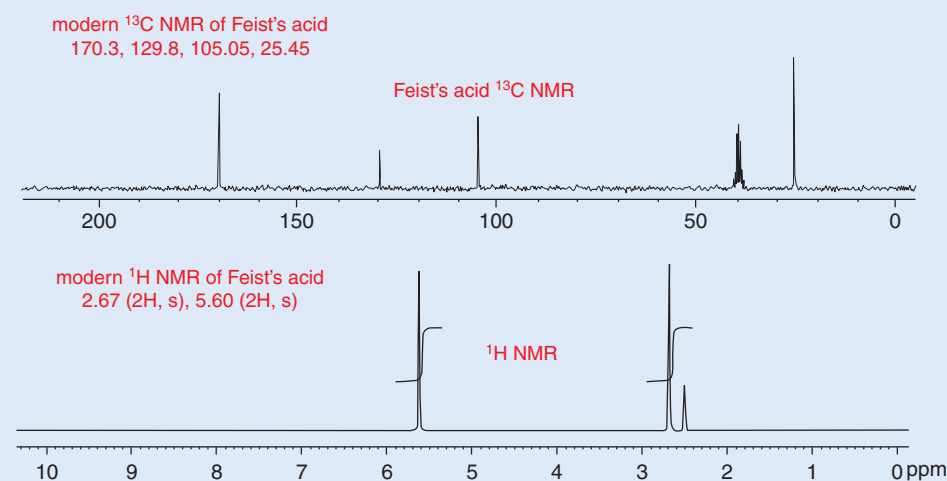
### The mystery of Feist's acid

It is hard nowadays to realize how difficult structure solving was before the days of spectroscopy. A celebrated case was that of 'Feist's acid', discovered by Feist in 1893 from a deceptively simple reaction. Early work without spectra led to two suggestions for its structure, both based on a three-membered ring, which gave the compound some fame because unsaturated three-membered rings were rare. The favoured structure was the cyclopropene.

The argument was still going on in the 1950s when the first NMR spectrometers appeared. Although infrared appeared to support the cyclopropene structure, one of the first problems resolved by the primitive 40 MHz instruments available was that of Feist's acid, which had no methyl group signal but did have two protons on a double bond and so had to be the exomethylene isomer after all.

This structure has two chiral centres, so how will we know which diastereoisomer we have? The answer was simple: the stereochemistry has to be *trans* because Feist's acid is chiral: it can be resolved (see later in this chapter) into two enantiomers. Now, the *cis* diacid would have a plane of symmetry, and so would be achiral—it would be a *meso* compound. The *trans* acid on the other hand is chiral. If you do not see this, try superimposing it on its mirror image—you will find that you cannot. In fact, Feist's acid has an *axis* of symmetry, and you will see shortly that axes of symmetry *are* compatible with chirality.

Modern NMR spectra make the structure easy to deduce. There are only two proton signals as the CO<sub>2</sub>H protons exchange in the DMSO solvent needed. The two protons on the double bond are identical (5.60 ppm) and so are the two protons on the three-membered ring, which come at the expected high field (2.67 ppm). There are four carbon signals: the C=O at 170 ppm, two alkene signals between 100 and 150 ppm, and the two identical carbons in the three-membered ring at 25.45 ppm.

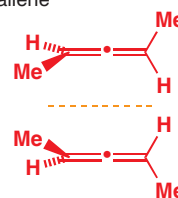


### Chiral compounds with no stereogenic centres

A few compounds are chiral, yet have no stereogenic centres. Try making a model of the allene in the margin. It has no stereogenic centre, but these mirror images are not superimposable and so the allene is chiral: the structures shown are enantiomers. Similarly, some biaryl compounds, such as the important bisphosphine below, known as BINAP, exist as two separate enantiomers because rotation about the green bond is restricted. If you were to look at this molecule straight down along the green bond, you would see that the two flat rings are at right angles to each other and so the molecule has a twist in it rather like the 90° twist in the allene. Compounds that are chiral because of restricted rotation about a single bond are called *atropisomers* (from the Greek for 'won't turn').

Interactive possible structures for Feist's acid

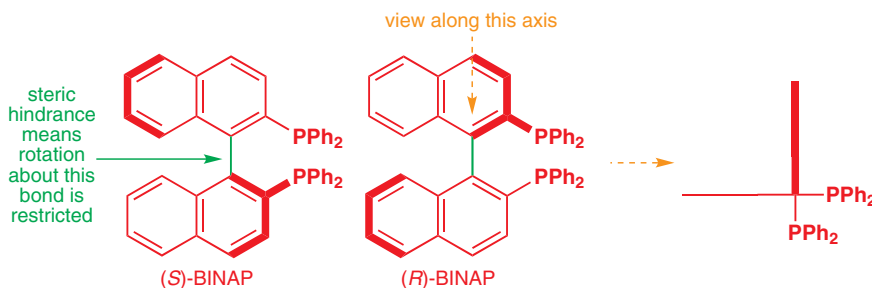
a chiral allene



Interactive chiral compounds without stereogenic centres—  
allene

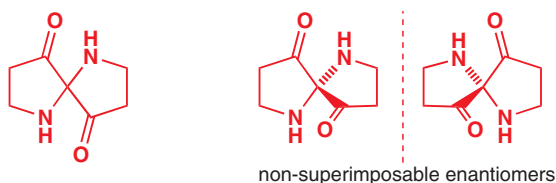
➡ We come back to BINAP in Chapter 41.

Interactive chiral compounds without stereogenic centres—BINAP

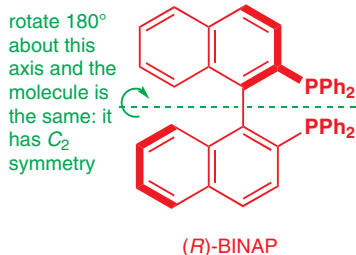


These two examples rely on the rigidity of  $\pi$  systems but this simple saturated system is also chiral. These two rings have to be orthogonal because of the tetrahedral nature of the central carbon atom. There is no chiral centre, but there is no plane of symmetry. Cyclic compounds like this with rings joined at a single C atom are called spiro compounds. Spiro compounds are often chiral even when at a first glance they look quite symmetrical, and you should look particularly carefully for planes of symmetry when you think about their stereochemistry.

Interactive chiral compounds without stereogenic centres—spiro amide



## Axes and centres of symmetry



Interactive BINAP showing  $C_2$  axis of symmetry

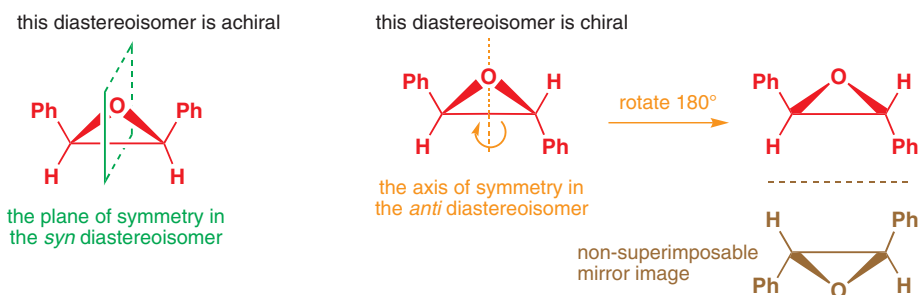
■ The subscript 2 means twofold axis of symmetry. Other orders of axial symmetry are possible in chemistry but are much rarer in simple organic compounds.

Interactive epoxide diastereoisomers showing plane of symmetry

■ We warned you that these statements (pp. 304 and 312) were incomplete: we are now about to complete them.

The fact that the three compounds we have just introduced (along with Feist's acid in the box on p. 319) were chiral might have surprised you, because at first glance they do look quite 'symmetrical'. In fact, they do all have an element of symmetry, and it is only one which is compatible with chirality: an *axis* of symmetry. If a molecule can be rotated through  $180^\circ$  about an axis to give exactly the same structure then it has twofold axial symmetry, or  $C_2$  symmetry. Compounds with an axis of symmetry will still be chiral, provided they lack either a plane or a centre of symmetry.

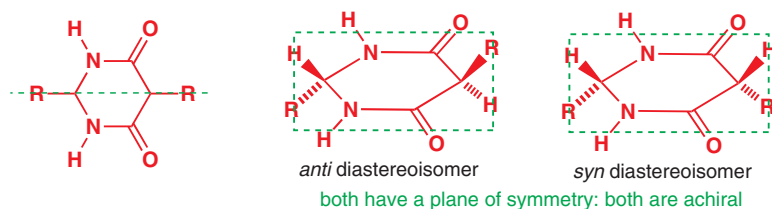
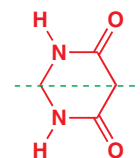
$C_2$  symmetry is common in many more everyday molecules than the ones in the last section. Below is an example of a compound with two diastereoisomers. One (we call it the *syn* diastereoisomer here—the two phenyl rings are on the same side) has a plane of symmetry—it must be achiral (and as it nonetheless has chiral centres we can also call it the *meso* diastereoisomer). The other has some degree of symmetry, but it has axial symmetry and can therefore be chiral. The  $C_2$  axis of symmetry is shown in orange. Rotating  $180^\circ$  gives back the same structure, but reflecting in a mirror plane (brown) gives a non-superimposable mirror image.



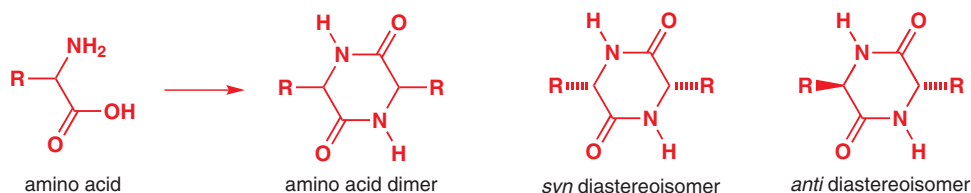
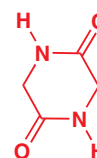
So far we have used a *plane of symmetry* as the defining characteristic of an achiral molecule: we have said several times that a molecule is chiral if it lacks a plane of symmetry. We are now

going to introduce a second type of symmetry that is not compatible with chirality. If a molecule has a *centre of symmetry* it is not chiral. We will now explain how to spot a centre of symmetry.

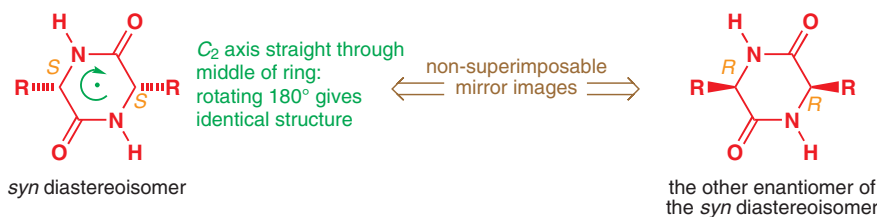
The diamide skeleton in the margin has a plane of symmetry in the plane of the page and also a plane of symmetry at right angles to that plane passing through the two saturated carbon atoms (represented by the green dotted line). If we add substituents R to this structure we can have two diastereoisomers with the two R groups on the same side (*syn*) of the flat ring or on opposite (*anti*) sides. Although the plane of the paper is no longer a plane of symmetry, neither isomer is chiral as the other plane bisects the substituents and is still a plane of symmetry. So far nothing new.



Now consider the related double amide below. The plane of the page is again a plane of symmetry but there is now no plane of symmetry at right angles. This heterocycle is called a 'diketopiperazine' and can be made by dimerizing an amino acid: the compound in the margin is the dimer of glycine. With substituted amino acids, such as those below where  $R \neq H$ , there are again two diastereoisomers, *syn* and *anti*. But their symmetry properties are different. The *syn* isomer is chiral but the *anti* isomer is not.

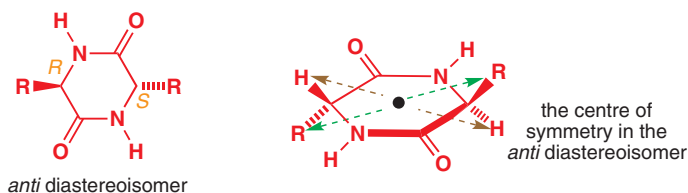


The *syn* diastereoisomer has no plane of symmetry but you should be able to spot a  $C_2$  axis of symmetry running straight through the middle of the ring. The axis is compatible with chirality of course. In this compound both chiral centres are *S* and it has an enantiomer where both are *R*.



Interactive diamides showing centre of symmetry

The *anti* diastereoisomer has no plane of symmetry, nor does it have an axis. Instead it has a *centre of symmetry*. This is marked with a black dot in the middle of the molecule and means that if you go in any direction from this centre and meet, say, an R group, you will meet the same thing if you go in the opposite direction (green arrows). The same thing applies to the brown arrows and, of course, to the ring itself. There is no centre of symmetry in the *syn* isomer as the green or brown arrows would point to R on one side and H on the other. The *anti* isomer is superimposable on its mirror image and is achiral.

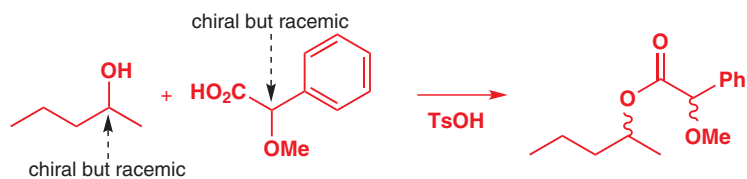


● **Chirality in terms of planes, centres, and axes of symmetry**

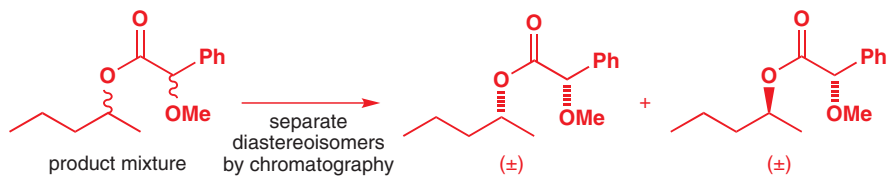
- Any molecule which has a *plane* of symmetry or a *centre* of symmetry is *achiral*.
- Any molecule which has an *axis* of symmetry is *chiral*, provided it does not also have a plane or a centre of symmetry. An axis of symmetry is the only symmetry element compatible with chirality.

## Separating enantiomers is called resolution

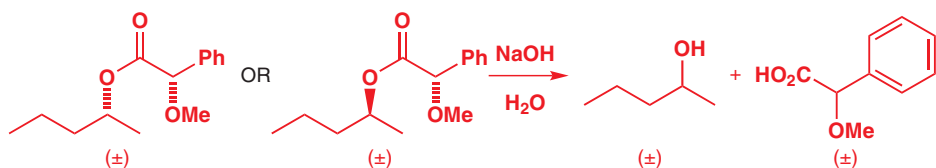
Early in this chapter we said that most of the molecules in Nature are chiral, and that Nature usually produces these molecules as single enantiomers. We've talked about the amino acids, the sugars, ephedrine, pseudoephedrine, and tartaric acid—all compounds that can be isolated from natural sources as single enantiomers. On the other hand, in the laboratory, if we make chiral compounds from achiral starting materials we are doomed to get racemic mixtures. So how do chemists ever isolate compounds as single enantiomers, other than by extracting them from natural sources? We'll consider this question in much more detail in Chapter 41, but here we will look at the simplest way: using Nature's enantiomerically pure compounds to help us separate the components of a racemic mixture into its two enantiomers. This process is called resolution. Imagine the reaction between a chiral, but racemic, alcohol and a chiral, but racemic, carboxylic acid, to give an ester in an ordinary acid-catalysed esterification (Chapter 10).



The product contains two chiral centres, so we expect to get two diastereoisomers, each a racemic mixture of two enantiomers. Diastereoisomers have different physical properties, so they should be easy to separate, for example by chromatography.

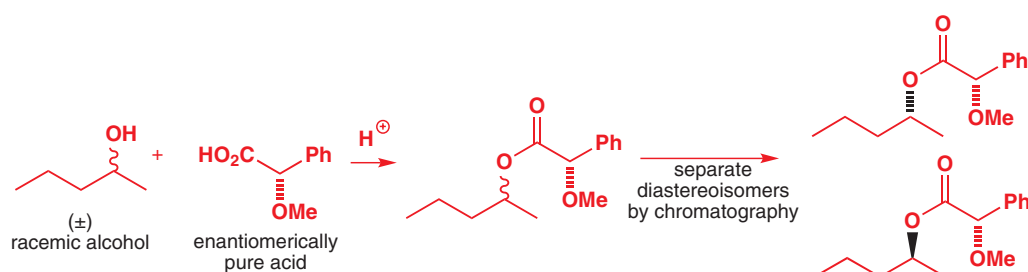


We could then reverse the esterification step and hydrolyse either of these diastereoisomers, to regenerate racemic alcohol and racemic acid.

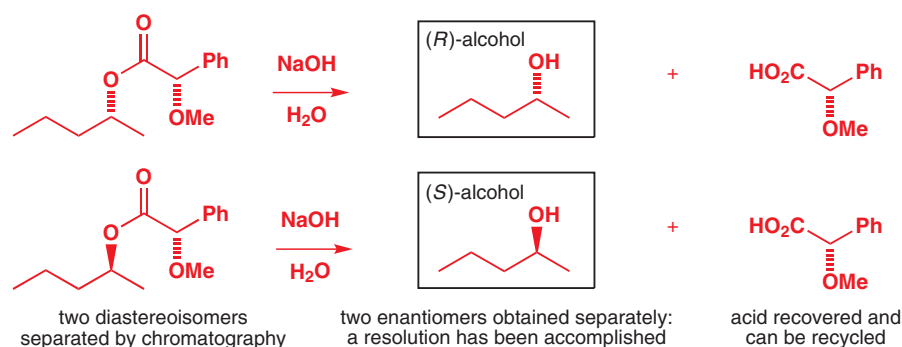


■ Remember that (±) means the compounds are racemic: we're showing only relative, not absolute, stereochemistry.

If we repeat this reaction, this time using an enantiomerically pure sample of the acid, available from (*R*)-mandelic acid, the almond extract you met on p. 310, we will again get two diastereoisomeric products, but this time each one will be enantiomerically pure. Note that the stereochemistry shown here *is* absolute stereochemistry.



If we now hydrolyse each diastereoisomer separately, we have done something rather remarkable: we have managed to separate two enantiomers of the starting alcohol.



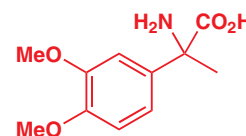
A separation of two enantiomers is called a **resolution**. Resolutions can be carried out only if we make use of a component that is already enantiomerically pure: it is very useful that Nature provides us with such compounds; resolutions nearly always make use of compounds derived from nature.

### Natural chirality

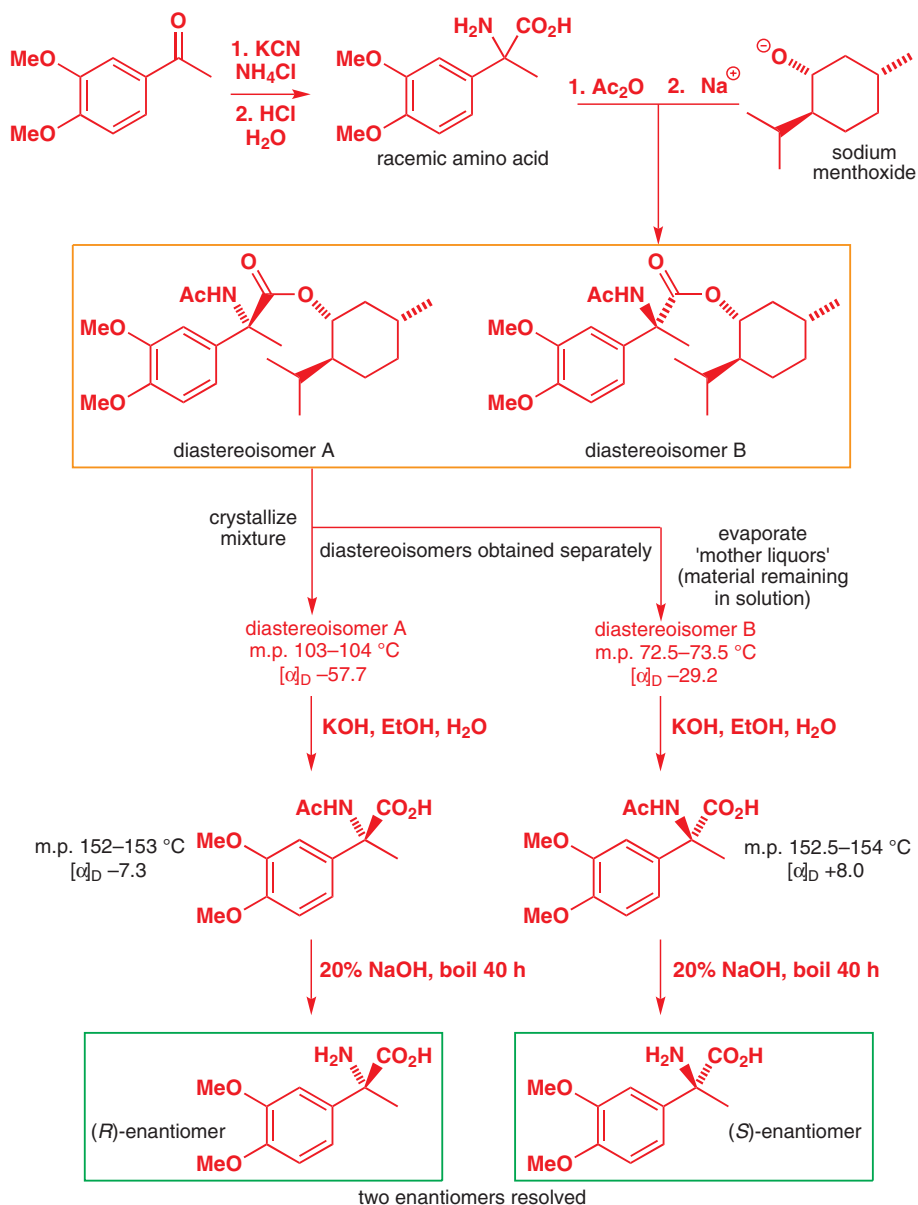
Why Nature uses only one enantiomer of most important biochemicals is an easier question to answer than how this asymmetry came about in the first place, or why L-amino acids and D-sugars were the favoured enantiomers, since, for example, proteins made out of racemic samples of amino acids would be complicated by the possibility of enormous numbers of diastereomers. Some have suggested that life arose on the surface of single chiral quartz crystals, which provided the asymmetric environment needed to make life's molecules enantiomerically pure. Or perhaps the asymmetry present in the spin of electrons released as gamma rays acted as a source of molecular asymmetry. Given that enantiomerically pure living systems should be simpler than racemic ones, maybe it was just chance that the L-amino acids and the D-sugars won out.

Now for a real example. Chemists studying the role of amino acids in brain function needed to obtain each of the two enantiomers of the amino acid in the margin. They made a racemic sample using the Strecker synthesis of amino acids that you met in Chapter 11. The racemic amino acid was treated with acetic anhydride to make the mixed anhydride and then with the sodium salt of naturally derived, enantiomerically pure alcohol menthol to give two diastereoisomers of the ester.

One of the diastereoisomers turned out to be more crystalline (that is, to have a higher melting point) than the other and, by allowing the mixture to crystallize, the chemists were able to isolate a pure sample of this diastereoisomer. Evaporating the diastereoisomer left in solution (the 'mother liquors') gave them the less crystalline diastereoisomer.



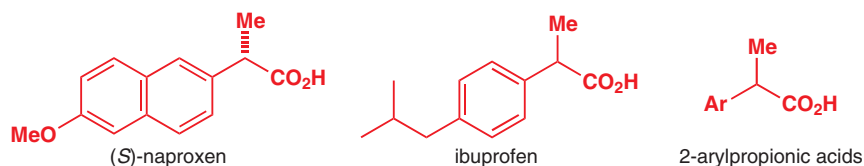
Next the esters were hydrolysed by boiling them in aqueous KOH. The acids obtained were enantiomers, as shown by their (nearly) opposite optical rotations and similar melting points. Finally, a more vigorous hydrolysis of the amides (boiling for 40 hours with 20% NaOH) gave them the amino acids they required for their biological studies (see bottom of p. 322).



### Resolutions using diastereoisomeric salts

The key point about resolution is that we must bring together two stereogenic centres in such a way that there is a degree of interaction between them: separable diastereoisomers are created from inseparable enantiomers. In the last two examples, the stereogenic centres were brought together in covalent compounds, esters. Ionic compounds will do just as well—in fact, they are often better because it is easier to recover the compound after the resolution.

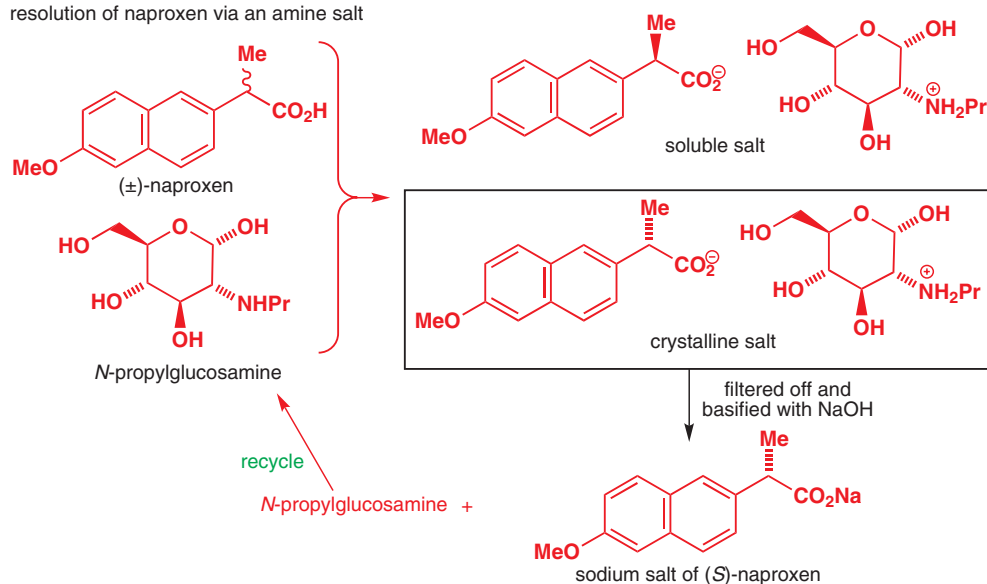
An important example is the resolution of the enantiomers of naproxen. Naproxen is a member of a family of compounds known as non-steroidal anti-inflammatory drugs (NSAIDs) which are 2-aryl propionic acids. This class also includes ibuprofen, the painkiller developed by Boots and marketed as Nurofen.



Both naproxen and ibuprofen are chiral but, while both enantiomers of ibuprofen are effective painkillers, and the drug is sold as a racemic mixture (and anyway racemizes in the body) only the (S) enantiomer of naproxen has anti-inflammatory activity. When the American pharmaceutical company Syntex first marketed the drug they needed a way of resolving the racemic naproxen they synthesized in the laboratory.

Since naproxen is a carboxylic acid, they chose to make the carboxylate salt of an enantiomerically pure amine, and found that the most effective was a glucose derivative. Crystals were formed, which consisted of the salt of the amine and (S)-naproxen, the salt of the amine with (R)-naproxen (the diastereoisomer of the crystalline salt) being more soluble and so remaining in solution. These crystals were filtered off and treated with base, releasing the amine (which can later be recovered and reused) and allowing the (S)-naproxen to crystallize as its sodium salt. This is an unusual resolving agent as a simpler amine might usually be preferred. However, it makes the point that many resolving agents may have to be tried before one is found that works.

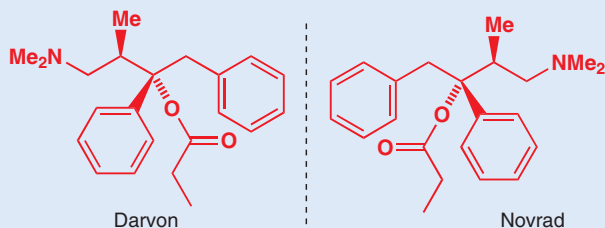
resolution of naproxen via an amine salt



### Chiral drugs

You may consider it strange that it was necessary to market naproxen as a single enantiomer, in view of what we have said about enantiomers having identical properties. The two enantiomers of naproxen do indeed have identical properties in the laboratory, but once they are inside a living system they, and any other chiral molecules, are differentiated by interactions with the enantiomerically pure molecules they find there. An analogy is that of a pair of gloves—the gloves weigh the same, are made of the same material, and have the same colour—in these respects they are identical. But interact them with a chiral environment, such as a hand, and they become differentiable because only one fits.

The way in which drugs interact with receptors mirrors this hand-and-glove analogy quite closely. Drug receptors, into which drug molecules fit like hands in gloves, are nearly always protein molecules, which are enantiomerically pure because they are made up of just L-amino acids. One enantiomer of a drug is likely to interact much better than the other, or perhaps in a different way altogether, so the two enantiomers of chiral drugs often have quite different pharmacological effects. In the case of naproxen, the (*S*)-enantiomer is 28 times as effective as the (*R*). Ibuprofen, on the other hand, is still marketed as a racemate because the compound racemizes in the bloodstream.

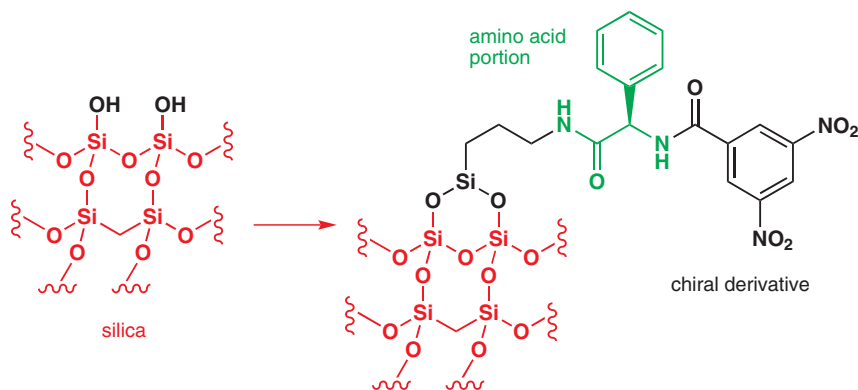


Sometimes, the enantiomers of a drug may have completely different therapeutic properties. One example is Darvon, which is a painkiller. Its enantiomer, known as Novrad, is an anticough agent. Notice how the enantiomeric relationship between these two drugs extends beyond their chemical structures! In Chapter 41 we will talk about other cases where two enantiomers have quite different biological effects.

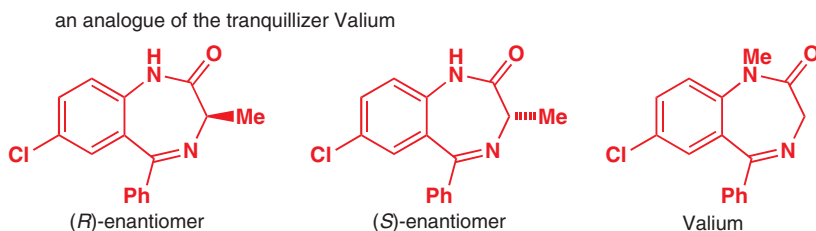
### Resolutions can be carried out by chromatography on chiral materials

■ Silica,  $\text{SiO}_2$ , is a macromolecular array of silicon and oxygen atoms. Its surface is covered with free OH groups, which can be used as an anchor for chiral derivatizing agents.

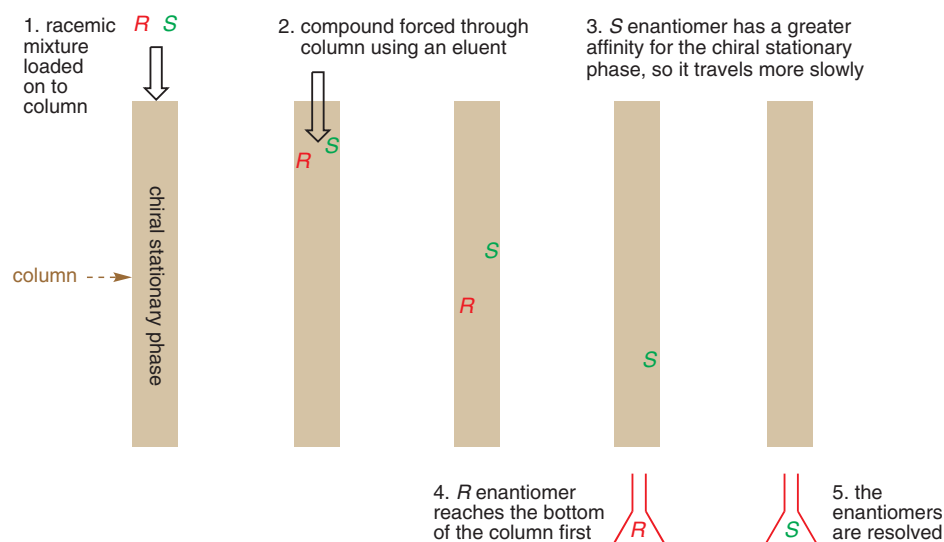
Interactions even weaker than ionic bonds can be used to separate enantiomers. Chromatographic separation relies on a difference in affinity between a stationary phase (often silica) and a mobile phase (the solvent travelling through the stationary phase, known as the eluent) mediated by, for example, hydrogen bonds or van der Waals interactions. If the stationary phase is made chiral by bonding it with an enantiomerically pure compound (often a derivative of an amino acid), chromatography can be used to separate enantiomers.



Chromatography on a chiral stationary phase is especially important when the compounds being resolved have no functional groups suitable for making the derivatives (usually esters or salts) needed for the more classical resolutions described above. For example, the two enantiomers of an analogue of the tranquillizer Valium were found to have quite different biological activities.



In order to study these compounds further, it was necessary to obtain them in enantiomerically pure form. This was done by passing a solution of the racemic compound through a column of silica bonded to an amino-acid-derived chiral stationary phase. The (*R*)-(-)-enantiomer showed a lower affinity for the stationary phase and therefore was eluted from the column first, followed by the (*S*)-(+)-enantiomer.



Two enantiomers of one molecule may be the same compound, but they are clearly different, although only in a limited number of situations. They can interact with biological systems differently, for example, and can form salts or compounds with different properties when reacted with a single enantiomer of another compound. In essence, enantiomers behave identically *except* when they are placed in a chiral environment. In Chapter 41 we will see how to use this fact to make single enantiomers of chiral compounds, but next we move on to three classes of reactions in which stereochemistry plays a key role: substitutions, eliminations, and additions.

■ You can think about chiral chromatography like this. Put yourself in this familiar situation: you want to help out a pensioner friend of yours who sadly lost his left leg in the war. A local shoe shop donates to you all their spare odd shoes, left and right, in his size (which happens to be the same as yours). You set about sorting the lefts from the rights, but are plunged into darkness by a power cut. What should you do? Well, you try every shoe on your right foot. If it fits you keep it; if not it's a left shoe and you throw it out.

Now this is just what chromatography on a chiral stationary phase is about. The stationary phase has lots of 'right feet' (one enantiomer of an adsorbed chiral molecule) sticking out of it, and as the mixture of enantiomers of 'shoes' flows past, 'right shoes' fit, and stick but 'left shoes' do not and flow on down the column, reaching the bottom first.

## Further reading

There are very many books on stereochemistry. The most comprehensive is probably E. L. Eliel and S. H. Wilen, *Stereochemistry of Organic Compounds*, Wiley Interscience, Chichester, 1994. But you may find this too comprehensive at this stage. A more accessible introduction is the Oxford Primer *Organic Stereochemistry*, M. J. T. Robinson, OUP, Oxford, 2001.

The first announcement of the correct structure of Feist's acid was by M. G. Ettinger, *J. Am. Chem. Soc.*, 1952, **74**, 5805 and an interesting follow-up article gives the NMR spectrum: W. E. von Doering and H. D. Roth, *Tetrahedron*, 1970, **26**, 2825.

## Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>